

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	2024	Date:	06/08/2026

Code of Conduct

I S.Venkateswarlu (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Venkateswarlu

QUESTIONS: 5

Total Marks: 50

Scenario-Based Assignment

Duration: 1hr

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question:

Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

ANSWER:

1. Machine Learning vs Deep Learning

Scenario

A healthcare company wants to detect pneumonia from chest X-ray images. Initially, they use Machine Learning (ML) with manually extracted features, but after collecting millions of images, they switch to Deep Learning (DL).

Comparison

Feature	Machine Learning	Deep Learning
Data Requirement	Works well with small and medium datasets.	Requires very large datasets (millions of images).
Feature Extraction	Features are manually designed (edges, texture, shape).	Automatically learns features directly from images.
Computational Resources	Runs on CPUs with lower memory requirements.	Requires GPUs/TPUs and high computational power.
Accuracy	Good for simple problems but limited for complex image analysis.	Higher accuracy for image classification and medical diagnosis.
Training Time	Faster training.	Longer training due to deep architectures.
Scalability	Performance improves slowly with more data.	Performance improves significantly with more data.

Why Deep Learning is Better

- Learns image features automatically without manual intervention.
- Handles millions of medical images efficiently.
- Detects complex disease patterns invisible to handcrafted features.
- Achieves higher diagnostic accuracy using Convolutional Neural Networks (CNNs).
- Continuously improves as more medical data becomes available.

Real-World Challenges

- Requires expensive GPU hardware.
- Needs large labeled datasets.
- Training takes considerable time.
- Medical models must be interpretable and reliable.
- Patient privacy and regulatory compliance must be ensured.

Conclusion

Deep Learning is the preferred choice because it automatically learns complex image features and delivers higher accuracy for medical image diagnosis when large datasets are available.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question:

Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex realworld datasets.

ANSWER:

2. Representation Learning

Definition

Representation Learning is a technique where a Deep Learning model automatically learns useful features from raw data instead of relying on manually engineered features.

Scenario

A retail company has millions of customer purchase records and wants to discover buying patterns automatically.

Working

Instead of manually selecting features such as:

- Total spending
- Number of purchases
- Favorite category

the neural network learns hidden purchasing behaviors directly from transaction data.

Traditional Feature Engineering vs Representation Learning

Traditional Feature Engineering	Representation Learning
Features created manually	Features learned automatically
Requires domain experts	Minimal human intervention
Limited feature quality	Learns highly informative features
Difficult for large datasets	Excellent for large datasets

Advantages

- Automatically discovers hidden customer behavior.
- Handles millions of transactions efficiently.
- Learns complex nonlinear relationships.
- Reduces manual effort.
- Improves recommendation systems and customer segmentation.

Applications

- Product recommendation
- Customer segmentation
- Fraud detection
- Demand forecasting

Conclusion

Representation Learning enables Deep Learning models to automatically discover meaningful patterns, making it ideal for analyzing large real-world retail datasets.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question:

Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

ANSWER:

Width and Depth of Neural Networks

Scenario

A startup wants to recognize handwritten digits.

Two proposals:

- Very wide network (one hidden layer with thousands of neurons)
- Deep network (many hidden layers with fewer neurons)

Width

- Number of neurons in one hidden layer.
- Wide networks can approximate many functions.
- Capture many features simultaneously.

Advantages

- Simpler architecture.
- Easier to train.
- Faster forward computation.

Disadvantages

- Requires many parameters.
- High memory usage.
- May overfit.
- Limited hierarchical feature learning.

Depth

Example:

Layer 1 → Edges

Layer 2 → Curves

Layer 3 → Shapes

Layer 4 → Digits

Comparison

Aspect	Wide Network	Deep Network
Feature Learning	Shallow	Hierarchical
Parameters	Very high	Moderate
Memory	High	Lower
Generalization	Poorer	Better
Accuracy	Moderate	Higher
Overfitting	More likely	Less likely (with regularization)

Recommendation

A deep neural network is more suitable because:

- Learns hierarchical image features.
- Requires fewer parameters.
- Provides better generalization.
- Achieves higher recognition accuracy.
- CNN-based deep architectures are the standard for handwritten digit recognition.

Conclusion

Deep networks outperform wide networks for image recognition tasks because they learn features progressively from simple to complex patterns.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question:

Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

ANSWER:

Activation Functions

Scenario

The model experiences:

- Slow convergence
- Vanishing gradients
- Dead neurons
- Unstable output layer

Comparison

Activation Function	Advantages	Disadvantages	Best Use
Sigmoid	Output between 0 and 1	Vanishing gradient	Binary output layer
Tanh	Zero-centered output	Still suffers vanishing gradients	Hidden layers (older models)
ReLU	Fast, simple, avoids vanishing gradients	Dead neurons	Hidden layers
Leaky ReLU	Prevents dead neurons	Small negative outputs	Hidden layers
ELU	Faster convergence, smoother learning	More computation	Deep hidden layers
Softmax	Converts outputs to probabilities	Only for classification output	Multi-class output layer

Recommended Usage

Layer	Activation
Hidden Layer 1	ReLU
Hidden Layer 2	Leaky ReLU
Hidden Layer 3	ELU (optional for deeper networks)

Layer	Activation
Binary Output Layer	Sigmoid
Multi-class Output Layer	Softmax

Justification

- ReLU speeds up training.
- Leaky ReLU prevents dead neurons.
- ELU improves convergence in deep networks.
- Sigmoid is suitable for binary classification.
- Softmax provides class probabilities for image classification.

Conclusion

Using ReLU/Leaky ReLU in hidden layers and Softmax in the output layer provides stable and efficient training for image classification models.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question:

Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

ANSWER:

Unsupervised Training of Neural Networks

Scenario

A social media platform collects billions of unlabeled user images.

Manual labeling is expensive and impractical.

Unsupervised Training

The neural network learns patterns directly from unlabeled data without requiring class labels.

Instead of learning:

Image → Cat

it learns:

- Shapes
- Textures
- Colors
- Objects
- Visual patterns

Learning Process

1. Input unlabeled images.

2. Learn hidden representations.
3. Discover similarities among images.
4. Build useful feature embeddings.
5. Use learned features for downstream tasks.

Advantages over Supervised Learning

Supervised Learning	Unsupervised Learning
Requires labeled data	No labels required
Labeling is expensive	Saves labeling cost
Limited by labeled samples	Uses all available data
Less scalable	Highly scalable

Suitable Techniques

1. Autoencoders
 - Learn compressed representations.
 - Useful for feature extraction and denoising.
2. Variational Autoencoders (VAEs)
 - Learn probabilistic latent representations.
 - Useful for representation learning and data generation.
3. Self-Supervised Learning
 - Creates learning tasks from the data itself.
 - Produces high-quality representations without manual labels.
4. Contrastive Learning
 - Learns by bringing similar images closer and pushing different images apart in the feature space.
5. Restricted Boltzmann Machines (RBMs)
 - Learn hidden feature representations from unlabeled data.

Benefits

- Eliminates manual labeling effort.
- Learns meaningful image features automatically.
- Scales to billions of images.
- Improves downstream tasks such as image search, recommendation, and classification after fine-tuning.

Conclusion

Unsupervised training enables neural networks to learn rich feature representations from massive unlabeled datasets, making it an effective and cost-efficient solution for large-scale social media image analysis.